

ZACHARY HILL

zacharyh728@gmail.com | +1 (732) 742-1299 | Boston, MA |  [ZacharyH728](#) |  [LinkedIn](#) |  [zhill.me](#)

EDUCATION

Northeastern University

Bachelor of Science in Electrical and Computer Engineering

Minor: Materials Science and Mathematics

Boston, MA | April 2027

GPA 3.3/4.0

WORK EXPERIENCE

DeLTA Labs at Northeastern University

July 2025 – Present

Research Assistant

Worked under Professor Alberto De la Torre projects include:

- Simulating and Optimizing 2.5 THz Distributed Bragg Reflector (DBR) microcavity using Tidy3D and their Finite-Difference Time-Domain (FDTD) simulation package. Also researched how to fabricate the simulated designs.
- Optimized and developed a Fourth Harmonic Generation System (FHG) setup to generate a 15mW UV beam from a 2.5W NIR laser
- Compared Sonnet's performance against Tidy3D's FDTD package in simulating a superconducting Grounded Co-Planar Waveguide (GCPW)

Northeastern University

Tutor for Khoury College of Computer Science

Spring 2024

- Tutored students in Fundamentals of Computer Science 2 using both Kotlin and Java

TECHNICAL SKILLS

Software: Tidy3D, AutoCAD, Fusion360, Linux, LTSpice, Proxmox, SolidWorks, and TrueNAS

Languages/Frameworks: Arduino, C, C++, CSS, HTML, Java, JavaScript, Node.js, React, and Typescript

Other: Nonlinear Optics, Laser Alignment, Laser Safety, Soldering, Laser-cutting, 3D-Printing (SLA/FDM), and Excel,

PROJECTS

Concrete Super-Capacitor (Group) – Northeastern University

Fall 2024

- Collaborated with group to research, design a custom-made super-capacitor using concrete as the main electrode
- Led designing and fabricating of the case and super-capacitor as well as the testing of its electrical properties.

RFID E.M.I.D. (Electromagnetic Interference Device)

October 2024

- A high frequency (13.56 MHz) active disruptor that produces a signal that blocks other RFID devices from receiving data.

16-Bit Calculator

January 2025

- Only using 2N2222 transistors, created a 16-bit calculator on a breadboard that is capable of addition, multiplication and subtraction.

Digital Wristwatch

December 2024

- Created a digital watch with a binary display that uses a microcontroller and a RTC to track time, in a custom 3D printed case.

Custom Lighting Console

Summer 2024

- Created various linear potentiometers, buttons, and rotary encoders are connected to an Arduino. All connected to a Raspberry Pi that hosts a web interface. Can interface with lights using the DMX-512 protocol, also capability of interfacing with ETC. consoles using OSC. Designed and built a custom 3D printed case in Fusion360.

EXTRACURRICULARS

FIRST Robotics Competition Team 25 Raider Robotics

Sept. 2019 – June 2023

Lead Electrician

2021-2023

- Led 5-7 high school students in designing and creating the electrical, motor and locomotive system for a robot each year to compete in FRC.

Lead Programmer

2021-2023

- Led 3-6 high school students in programming the robot using and team utility apps such as a Robot Tracking/Scouting Android app. Programing languages involved include: Java, C++, Kotlin

SKILLS AND CERTIFICATES

- German Proficiency: A2
- First Aid Certified: Certified by the American Red Cross in First Aid, CPR, and AED
- Boy Scouts of America: Eagle Scout

INTRESTS

Videogames, Hiking, Camping, 3D printing, Home-lab/Self-Hosting, Oboe, Reading, Lighting Design
